

Mark Scheme

Q1.

Question	Scheme	Marks	AOs
(a)	$x^2 + 8x - 3 = (Ax + B)(x + 2) + C$ or $Ax(x + 2) + B(x + 2) + C$ $\Rightarrow A = \dots, B = \dots, C = \dots$ or $\begin{array}{r} x+6 \\ x+2 \overline{)x^2+8x-3} \\ \underline{x^2+2x} \\ 6x-3 \\ \underline{6x+12} \\ -15 \end{array}$	M1	1.1b
	Two of $A = 1, B = 6, C = -15$	A1	1.1b
	All three of $A = 1, B = 6, C = -15$	A1	1.1b
	(3)		
6(b)	$\int \frac{x^2 + 8x - 3}{x + 2} dx = \int x + 6 - \frac{15}{x + 2} dx = \dots - 15 \ln(x + 2)$	M1	1.1b
	$= \frac{1}{2}x^2 + 6x - 15 \ln(x + 2) \quad (+c)$	A1ft	1.1b
	$\int_0^6 \frac{x^2 + 8x - 3}{x + 2} dx = \left[\frac{1}{2}x^2 + 6x - 15 \ln(x + 2) \right]_0^6$ $= (18 + 36 - 15 \ln 8) - (0 + 0 - 15 \ln 2)$ $= 18 + 36 - (15 - 45) \ln 2$ or e.g. $18 + 36 + 15 \ln\left(\frac{2}{8}\right)$	M1	2.1
	$= 54 - 30 \ln 2$	A1	1.1b
	(4)		
(7 marks)			

Notes:

- (a)
- M1:** Multiplies by $(x + 2)$ and attempts to find values for A, B and C e.g. by comparing coefficients or substituting values for x . If the method is unclear, at least 2 terms must be correct on rhs.
 Or attempts to divide $x^2 + 8x - 3$ by $x + 2$ and obtains a linear quotient and a constant remainder.
 This mark may be implied by 2 correct values for A, B or C
- A1:** Two of $A = 1, B = 6, C = -15$. But note that just performing the division correctly is insufficient and they must clearly identify their A, B, C to score any accuracy marks.
- A1:** All three of $A = 1, B = 6, C = -15$
 This is implied by stating $\frac{x^2 + 8x - 3}{x + 2} = x + 6 - \frac{15}{x + 2}$ or within the integral in (b)
- (b)
- M1:** Integrates an expression of the form $\frac{C}{x + 2}$ to obtain $k \ln(x + 2)$.
 Condone the omission of brackets around the " $x + 2$ "
- A1ft:** Correct integration ft on their $Ax + B + \frac{C}{x + 2}$, ($A, B, C \neq 0$) The brackets should be present around the " $x + 2$ " unless they are implied by subsequent work.
- M1:** Substitutes both limits 0 and 6 into an expression that contains an x or x^2 term or both and a $\ln$ term and subtracts either way round **WITH** fully correct log work to combine two log terms (but allow sign errors when removing brackets) leading to an answer of the form $a + b \ln c$ (a, b and c not necessarily integers)
 e.g. if they expand to get $-15 \ln 8 - 15 \ln 2$ followed by $-15 \ln 16$ and reach $a + b \ln c$ then allow the M mark
- A1:** $54 - 30 \ln 2$ (Apply isw once a correct answer is seen)

Q2.

Question	Scheme	Marks	AOs
(a)	$x = u^2 + 1 \Rightarrow dx = 2u du$ oe	B1	1.1b
	Full substitution $\int \frac{3dx}{(x-1)(3+2\sqrt{x-1})} = \int \frac{3 \times 2u du}{(u^2+1-1)(3+2u)}$	M1	1.1b
	Finds correct limits e.g. $p = 2, q = 3$	B1	1.1b
	$= \int \frac{3 \times 2u' du}{u^2(3+2u)} = \int \frac{6 du}{u(3+2u)} *$	A1*	2.1
		(4)	
(b)	$\frac{6}{u(3+2u)} = \frac{A}{u} + \frac{B}{3+2u} \Rightarrow A = \dots, B = \dots$	M1	1.1b
	Correct PF. $\frac{6}{u(3+2u)} = \frac{2}{u} - \frac{4}{3+2u}$	A1	1.1b
	$\int \frac{6 du}{u(3+2u)} = 2 \ln u - 2 \ln(3+2u) \quad (+c)$	dM1 A1ft	3.1a 1.1b
	Uses limits $u = "3", u = "2"$ with some correct $\ln$ work leading to $k \ln b$ E.g. $(2 \ln 3 - 2 \ln 9) - (2 \ln 2 - 2 \ln 7) = 2 \ln \frac{7}{6}$	M1	1.1b
	$\ln \frac{49}{36}$	A1	2.1
		(6)	
(10 marks)			
Notes: Mark (a) and (b) together as one complete question			

(a)

B1: $dx = 2u du$ or exact equivalent. E.g. $\frac{dx}{du} = 2u, \frac{du}{dx} = \frac{1}{2}(x-1)^{-\frac{1}{2}}$

M1: Attempts a full substitution of $x = u^2 + 1$, including $dx \rightarrow 2u du$ to form an integrand in terms of u .
Condone slips but there should be an attempt to use the correct substitution on the denominator.

B1: Finds correct limits either states $p = 2, q = 3$ or sight of embedded values as limits to the integral

A1*: Clear reasoning including one fully correct intermediate line, including the integral signs, leading to the given expression ignoring limits. So B1, M1, B0, A1 is possible if the limits are incorrect, omitted or left as 5 and 10.

(b)

M1: Uses correct form of PF leading to values of A and B .

A1: Correct PF $\frac{6}{u(3+2u)} = \frac{2}{u} - \frac{4}{3+2u}$ (Not scored for just the correct values of A and B)

dM1: This is an overall problem solving mark. It is for using the correct PF form and integrating using $\ln$ s.
Look for $P \ln u + Q \ln(3+2u)$

A1ft: Correct integration for their $\frac{A}{u} + \frac{B}{3+2u} \rightarrow A \ln u + \frac{B}{2} \ln(3+2u)$ with or without modulus signs

M1: Uses their 2 and 3 as limits, with at least one correct application of the addition law or subtraction law leading to the form $k \ln b$ or $\ln a$. PF's must have been attempted. Condone bracketing slips. Alternatively changing the u 's back to x 's and use limits of 5 and 10.

A1: Proceeds to $\ln \frac{49}{36}$. Answers without working please send to review.

Q3.

Question	Scheme	Marks	AOs
(a)	$N = aT^b \Rightarrow \log_{10} N = \log_{10} a + \log_{10} T^b$	M1	2.1
	$\Rightarrow \log_{10} N = \log_{10} a + b \log_{10} T$ so $m = b$ and $c = \log_{10} a$	A1	1.1b
		(2)	
(b)	Uses the graph to find either a or b $a = 10^{\text{intercept}}$ or $b = \text{gradient}$	M1	3.1b
	Uses the graph to find both a and b $a = 10^{\text{intercept}}$ and $b = \text{gradient}$	M1	1.1b
	Uses $T = 3$ in $N = aT^b$ with their a and b	M1	3.1b
	Number of microbes ≈ 800	A1	1.1b
		(4)	
(c)	$N = 1000000 \Rightarrow \log_{10} N = 6$	M1	3.4
	We cannot 'extrapolate' the graph and assume that the model still holds	A1	3.5b
		(2)	
(d)	States that ' a ' is the number of microbes 1 day after the start of the experiment	B1	3.2a
		(1)	
(9 marks)			

Notes:

(a)

M1: Takes logs of both sides and shows the addition law

M1: Uses the power law, writes $\log_{10} N = \log_{10} a + b \log_{10} T$ and states $m = b$ and $c = \log_{10} a$

(b)

M1: Uses the graph to find either a or b $a = 10^{\text{intercept}}$ or $b = \text{gradient}$. This would be implied by the sight of $b = 2.3$ or $a = 10^{1.8} \approx 63$

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M1: Uses $T = 3 \Rightarrow N = aT^b$ with their a and b . This is implied by an attempt at $63 \times 3^{2.3}$

A1: Accept a number of microbes that are approximately 800. Allow 800 ± 150 following correct work.

There is an alternative to this using a graphical approach.

M1: Finds the value of $\log_{10} T$ from $T = 3$. Accept as $T = 3 \Rightarrow \log_{10} T \approx 0.48$

M1: Then using the line of best fit finds the value of $\log_{10} N$ from their "0.48"
Accept $\log_{10} N \approx 2.9$

M1: Finds the value of N from their value of $\log_{10} N$ $\log_{10} N \approx 2.9 \Rightarrow N = 10^{2.9}$

A1: Accept a number of microbes that are approximately 800. Allow 800 ± 150 following correct work

(c)

M1 For using $N = 1000000$ and stating that $\log_{10} N = 6$

A1: Statement to the effect that "we only have information for values of $\log N$ between 1.8 and 4.5 so we cannot be certain that the relationship still holds". "We cannot extrapolate with any certainty, we could only interpolate"

There is an alternative approach that uses the formula.

M1: Use $N = 1000000$ in their $N = 63 \times T^{2.3} \Rightarrow \log_{10} T = \frac{\log_{10} \left(\frac{1000000}{63} \right)}{2.3} \approx 1.83$.

A1: The reason would be similar to the main scheme as we only have $\log_{10} T$ values from 0 to 1.2. We cannot 'extrapolate' the graph and assume that the model still holds

(d)

B1: Allow a numerical explanation $T = 1 \Rightarrow N = a1^b \Rightarrow N = a$ giving a is the value of N at $T = 1$

Q4.

Question	Scheme	Marks	AOs
	$4^{3p-1} = 5^{210} \Rightarrow (3p-1)\log 4 = 210\log 5$	M1	1.1b
	$\Rightarrow 3p = \frac{210\log 5}{\log 4} + 1 \Rightarrow p = \dots$	dM1	2.1
	$p = \text{awrt } 81.6$	A1	1.1b
		(3)	
(3 marks)			
Notes:			

M1: Takes logs of both sides and uses the power law on each side.

Condone a missing bracket on lhs and slips.

Award for any base including ln but the logs must be the same base.

dM1: A full method leading to a value for p .

It is dependent upon the previous M mark and there must be an attempt to change the subject of the equation in the correct order.

Look for $(3p-1)\log 4 = 210\log 5 \Rightarrow 3p = \frac{210\log 5}{\log 4} \pm 1 \Rightarrow p = \dots$ condoning slips.

You may see numerical versions E.g. $(3p-1) \times 0.60 = 210 \times 0.7 \Rightarrow 1.8p - 0.6 = 147 \Rightarrow p = 82$

Use of incorrect log laws would be dM0. E.g. $(3p-1)\log 4 = 210\log 5 \Rightarrow 3p = 210\log \frac{5}{4} \pm 1$

A1: awrt 81.6 following a correct method. Bracketing errors can be recovered for full marks

A correct answer with no working scores 0 marks. The demand in the question is clear.

There are alternatives:

E.g. A starting point could be $4^{3p-1} = 5^{210} \Rightarrow \frac{4^{3p}}{4} = 5^{210}$

but the first M mark is still for using the power law correctly on each side

In such a method the dM1 mark is for using **all** log rules correctly and proceeding to a value for p .

Using base 4 or 5

E.g. $4^{3p-1} = 5^{210} \Rightarrow (3p-1) = \log_4 5^{210}$

The M mark is not scored until $(3p-1) = 210\log_4 5$

Question	Scheme	Marks	AOs
(a)	$2\log(4-x) = \log(4-x)^2$	B1	1.2
	$2\log(4-x) = \log(x+8) \Rightarrow \log(4-x)^2 = \log(x+8)$ $(4-x)^2 = (x+8)$ or $2\log(4-x) = \log(x+8) \Rightarrow \log(4-x)^2 - \log(x+8) = 0$ $\frac{(4-x)^2}{(x+8)} = 1$	M1	1.1b
	$16 - 8x + x^2 = x + 8 \Rightarrow x^2 - 9x + 8 = 0 *$	A1*	2.1
	(3)		
	(a) Alternative - working backwards:		
	$x^2 - 9x + 8 = 0 \Rightarrow (4-x)^2 - x - 8 = 0$	B1	1.2
	$\Rightarrow (4-x)^2 = x + 8$ $\Rightarrow \log(4-x)^2 = \log(x+8)$	M1	1.1b
	$\Rightarrow 2\log(4-x) = \log(x+8) *$ Hence proved.	A1	2.1
(b)	(i) $(x =) 1, 8$	B1	1.1b
	(ii) 8 is not a solution as $\log(4-8)$ cannot be found	B1	2.3
	(2)		
(5 marks)			

Notes:

(a)

B1: States or uses $2\log(4-x) = \log(4-x)^2$ **M1:** Correct attempt at eliminating the logs to form a quadratic equation in x .

Note that this may be implied by e.g. $\log \frac{(4-x)^2}{(x+8)} = 0 \Rightarrow (4-x)^2 = x+8$

A1*: Proceeds to the given answer with at least one line where the $(4-x)^2$ has been multiplied out.

There must be no errors or omissions but condone invisible brackets around the arguments of the logs e.g. allow $\log 16 - 8x + x^2$ for $\log(16 - 8x + x^2)$ and $\log x + 8$ for $\log(x+8)$

Note we will allow a start of $(4-x)^2 = x+8$ with no previous work for full marks.

Some examples of how to mark (a) in particular cases:

$$2\log(4-x) = \log(x+8) \Rightarrow \log(4-x)^2 = \log(x+8) \Rightarrow \frac{\log(4-x)^2}{\log(x+8)} = 1$$

$$2\log(4-x) = \log(x+8) \Rightarrow \log(4-x)^2 - \log(x+8) = 0 \Rightarrow (4-x)^2 - x - 8 = 0$$

$$\Rightarrow 16 - 8x + x^2 - x - 8 \Rightarrow x^2 - 9x + 8 = 0$$

Scores B1M1A1

$$2\log(4-x) = \log(x+8) \Rightarrow \log(4-x)^2 - \log(x+8) = 0 \Rightarrow \frac{\log(4-x)^2}{\log(x+8)} = 0$$

$$\Rightarrow \frac{(4-x)^2}{(x+8)} = 1 \Rightarrow 16 - 8x + x^2 = x + 8 \Rightarrow x^2 - 9x + 8 = 0$$

Scores B1M0A0

(a) Alternative:

B1: Writes $x^2 - 9x + 8 = 0$ as $(4 - x)^2 - x - 8 = 0$ or equivalent

M1: Proceeds correctly to reach $\log(4 - x)^2 = \log(x + 8)$

A1: Obtains $2\log(4 - x) = \log(x + 8)$ and makes a (minimal) conclusion e.g. hence proved, QED, #, square etc.

(b)

B1: Writes down $(x =) 1, 8$

B1: Chooses 8 (no follow through here) and gives a reason why it should be rejected by referring to logs and which log it is.

They must refer to the 8 as the required value but allow e.g. $x \neq 8$ and there must be a reference to $\log(4 - x)$ or log of lhs or $\log(-4)$ or the $4 - 8$. Some acceptable reasons are: $\log(-4)$ can't be found/worked out/is undefined, $\log(-4)$ gives math error, $\log(-4) = \text{n/a}$, lhs is $\log(\text{negative})$ so reject, you can't do the log of a negative number which would happen with $4 - 8$

Do not allow "you can't have a negative log" unless this is clarified further and do not allow "you get a math error" in isolation

There must be no contradictory statements.

Note that this is an independent mark but must have $x = 8$ (i.e. may have solved to get $x = -1, 8$ for first B mark)